EXPERIMENT-8

DATA PRE-PROCESSING & FEATURE ENGINEERING

AIM:

To implement data pre-processing and feature engineering techniques on a dataset including removing unnecessary features, feature scaling and encoding categorical variables using Python libraries

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. A dataset for Practice
3. Jupyter Notebook
4. Knowledge on Pandas & Scikit-Learn Libraries

Step 1: Import the basic libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load & Analyze the Dataset

In [ ]:
```
# Load the student dataset
df_241UDS0092 = pd.read_excel("/content/research_student (1).xlsx")
```

In [ ]:
```
#Dataset Information
df_241UDS0092.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 223 entries, 0 to 222
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Branch                  220 non-null    object 
 1   Marks[10th]             220 non-null    float64
 2   Marks[12th]             220 non-null    float64
 3   Gender                  220 non-null    object 
 4   Board[10th]             220 non-null    object 
 5   Board[12th]             220 non-null    object 
 6   Category                220 non-null    object 
 7   GPA 1                   220 non-null    float64
 8   Rank                    221 non-null    float64
 9   Normalized Rank         220 non-null    float64
 10  CGPA                    220 non-null    float64
 11  Current Back            220 non-null    float64
 12  Ever Back               219 non-null    float64
 13  GPA 2                   220 non-null    float64
 14  GPA 3                   220 non-null    float64
 15  GPA 4                   220 non-null    float64
 16  GPA 5                   220 non-null    float64
 17  GPA 6                   220 non-null    float64
 18  Olympiads Qualified     220 non-null    float64
 19  Technical Projects      220 non-null    float64
 20  Tech Quiz               220 non-null    float64
 21  Engg. Coaching          220 non-null    float64
 22  NTSE Scholarships       220 non-null    float64
 23  Miscellany Tech Events  220 non-null    float64
dtypes: float64(19), object(5)
memory usage: 41.9+ KB
```

In [ ]:
```
df_241UDS0092.head()
```

Out[5]:

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	CIVIL	77.57	64.6	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	86.40	71.8	Male	CBSE	CBSE	GEN
3	CSE	88.14	78.0	Male	ICSE	ICSE	GEN
4	CSE	65.40	59.8	Female	CBSE	CBSE	ST

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	NaN	NaN	NaN	...	NaN	NaN	NaN
1	6.29	44718.0	15.970714	...	5.94	5.41	6.25
2	6.47	24222.0	8.650714	...	5.88	5.53	6.44
3	7.35	24723.0	8.829643	...	6.54	6.41	6.50
4	6.41	232157.0	82.913214	...	5.71	5.24	5.88

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0
4	6.25	2.0	0.0	2.0	2.0	0.0	1.0

5 rows × 24 columns

In [ ]:
```
#Check the shape
df_241UDS0092.shape
```

Out[6]:
```
(223, 24)
```

In [ ]:
```
#Check for null values
df_241UDS0092.isnull().sum()
```

Out[7]:

	0
Branch	3
Marks[10th]	3
Marks[12th]	3
Gender	3
Board[10th]	3
Board[12th]	3
Category	3
GPA 1	3
Rank	2
Normalized Rank	3
CGPA	3
Current Back	3
Ever Back	4
GPA 2	3
GPA 3	3
GPA 4	3
GPA 5	3
GPA 6	3
Olympiads Qualified	3
Technical Projects	3
Tech Quiz	3
Engg. Coaching	3
NTSE Scholarships	3
Miscellany Tech Events	3

dtype: int64

In [ ]:
```
df_241UDS0092 = df_241UDS0092.dropna(how='all')

df_241UDS0092.shape
```

Out[8]:
```
(221, 24)
```

Step 3: Remove Unnecessary Features

In [ ]:
```
# Removing unnecessary Rank column
df_241UDS0092 = df_241UDS0092.drop(
    'Rank',
    axis=1
)

df_241UDS0092.shape
```

Out[9]:
```
(221, 23)
```

In [ ]:
```
df_241UDS0092.columns
```

Out[10]:
```
Index(['Branch', 'Marks[10th]', 'Marks[12th]', 'Gender', 'Board[10th]',
       'Board[12th]', 'Category', 'GPA 1', 'Normalized Rank', 'CGPA',
       'Current Back', 'Ever Back', 'GPA 2', 'GPA 3', 'GPA 4', 'GPA 5',
       'GPA 6', 'Olympiads Qualified', 'Technical Projects', 'Tech Quiz',
       'Engg. Coaching', 'NTSE Scholarships', 'Miscellany Tech Events'],
      dtype='object')
```

Step 4: Scale the Numerical Features

In [11]:
```
from sklearn.preprocessing import StandardScaler
```

In [ ]:
```
#Create an instance
z_scaler_241UDS0092 = StandardScaler()
```

In [ ]:
```
#Scale the numerical features
df_241UDS0092[
    ['Marks[10th]', 'Marks[12th]']
] = z_scaler_241UDS0092.fit_transform(
    df_241UDS0092[['Marks[10th]', 'Marks[12th]']]
)
```

In [ ]:
```
#After Scaling
df_241UDS0092.head(3)
```

Out[14]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
1	CIVIL	-0.792630	-1.460973	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	0.246178	-0.663065	Male	CBSE	CBSE	GEN
3	CSE	0.450881	0.024023	Male	ICSE	ICSE	GEN

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
1	6.29	15.970714	6.02	...	5.94	5.41	6.25
2	6.47	8.650714	6.10	...	5.88	5.53	6.44
3	7.35	8.829643	6.65	...	6.54	6.41	6.50

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0

3 rows × 23 columns

Apply Normalization using MinMaxScaler

In [17]:
```
#Import MinMaxScaler
from sklearn.preprocessing import MinMaxScaler
```

In [18]:
```
#Create an instance
mm_scaler_241UDS0100 = MinMaxScaler()
```

In [ ]:
```
#Scale the numerical features
df_241UDS0092[
    ['CGPA', 'GPA 1']
] = mm_scaler_241UDS0092.fit_transform(
    df_241UDS0092[['CGPA', 'GPA 1']]
)
```

In [ ]:
```
#After Scaling
df_241UDS0092.head(3)
```

Out[20]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
1	CIVIL	-0.792630	-1.460973	Male	BSEB Patna	BSEB Patna	OBC
2	CSE	0.246178	-0.663065	Male	CBSE	CBSE	GEN
3	CSE	0.450881	0.024023	Male	ICSE	ICSE	GEN

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
1	0.007688	15.970714	0.041667	...	5.94	5.41	6.25
2	0.010299	8.650714	0.067308	...	5.88	5.53	6.44
3	0.023064	8.829643	0.243590	...	6.54	6.41	6.50

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0

3 rows × 23 columns

Step 5: Encode the Categorical Features

Apply Integer Encoding using LabelEncoder

In [21]:
```
from sklearn.preprocessing import LabelEncoder
```

In [ ]:
```
encoder_241UDS0092 = LabelEncoder()
```

In [ ]:
```
df_241UDS0092['Gender'] = encoder_241UDS0092.fit_transform(
    df_241UDS0092['Gender']
)
```

In [ ]:
```
df_241UDS0092['Category'] = encoder_241UDS0092.fit_transform(
    df_241UDS0092['Category']
)
```

In [ ]:
```
df_241UDS0092['Branch'] = encoder_241UDS0092.fit_transform(
    df_241UDS0092['Branch']
)
```

In [ ]:
```
df_241UDS0092.head(3)
```

Out[28]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
1	0	-0.792630	-1.460973	1	BSEB Patna	BSEB Patna	1
2	1	0.246178	-0.663065	1	CBSE	CBSE	0
3	1	0.450881	0.024023	1	ICSE	ICSE	0

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
1	0.007688	15.970714	0.041667	...	5.94	5.41	6.25
2	0.010299	8.650714	0.067308	...	5.88	5.53	6.44
3	0.023064	8.829643	0.243590	...	6.54	6.41	6.50

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
1	6.13	1.0	4.0	3.0	2.0	0.0	5.0
2	6.19	2.0	2.0	0.0	1.0	0.0	4.0
3	6.69	1.0	1.0	3.0	0.0	0.0	5.0

3 rows × 23 columns

Apply One-Hot Encoding using pd.get_dummies()

In [ ]:
```
df_241UDS0092 = pd.get_dummies(
    df_241UDS0092,
    columns=['Board[10th]', 'Board[12th]'],
    drop_first=True
)
```

In [ ]:
```
df_241UDS0092.head(3)
```

Out[30]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Category	GPA 1	Normalized Rank
1	0	-0.792630	-1.460973	1	1	0.007688	15.970714
2	1	0.246178	-0.663065	1	0	0.010299	8.650714
3	1	0.450881	0.024023	1	0	0.023064	8.829643

Columns 9–15 of 22 · first column repeated

	CGPA	Current Back	Ever Back	...	Board[12th]_RBSE	Board[12th]_STATE BOARD	Board[12th]_STATE BOARD OF ANDHRA PRADESH
1	0.041667	1.0	4.0	...	False	False	False
2	0.067308	1.0	7.0	...	False	False	False
3	0.243590	1.0	1.0	...	False	False	False

Columns 16–22 of 22 · first column repeated

	Board[12th]_U.P. BOARD	Board[12th]_UP BOARD	Board[12th]_bhopal	Board[12th]_bihar school examination board	Board[12th]_c.b.s.e	Board[12th]_cbse	Board[12th]_state
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False

3 rows × 106 columns

In [ ]:
```
df_241UDS0092.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 221 entries, 1 to 222
Columns: 106 entries, Branch to Board[12th]_state
dtypes: bool(85), float64(18), int64(3)
memory usage: 56.3 KB
```

Convert Boolean Datatypes into Integers

In [ ]:
```
bool_columns_241UDS0092 = df_241UDS0092.select_dtypes(
    include='bool'
).columns
```

In [ ]:
```
for col in bool_columns_241UDS0092:
    df_241UDS0092[col] = df_241UDS0092[col].astype(int)
```

In [ ]:
```
df_241UDS0092.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 221 entries, 1 to 222
Columns: 106 entries, Branch to Board[12th]_state
dtypes: float64(18), int64(88)
memory usage: 184.7 KB
```

In [ ]:
```
df_241UDS0092.shape
```

Out[35]: (221, 106)

In [ ]:
```
df_241UDS0092.to_csv(
    'processed_student_dataset_241UDS0092.csv',
    index=False
)
```

RESULT:

The student dataset was pre-processed by removing an unnecessary feature, scaling numerical features using StandardScaler and MinMaxScaler, and encoding categorical variables using Label Encoding and One-Hot Encoding. The dataset is now transformed into a clean numerical form, ready for use in machine learning models.